

REVIEW

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Systematised evidence mapping of generative artificial intelligence (GenAI) and digital divide phenomena in higher education

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Abstract

This systematic evidence mapping study maps and synthesises a small but growing body of empirical work exploring the crossroads between Generative Artificial Intelligence (GenAI) technologies and digital equity concerns in diverse higher education settings. A comprehensive three-stage search methodology, following systematic identification procedures aligned with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-ScR) standards, accessed and evaluated twelve included publications from 1 January 2023 to 30 January 2025 (including online-first articles available by that date). The review extends longstanding digital divide scholarship by focusing specifically on GenAI-related configurations of access, skills, and outcomes in the post-2022 period. Additional contextualisation was provided through selective analysis of the Trust in AI Report 2025 (University of Melbourne & KPMG), which surveyed public attitudes towards AI across 47 countries and examined how employees and students use and experience AI in work and education settings, and is treated here as sector-relevant background survey evidence rather than as an additional empirical strand of the review. Analysis reveals that whilst GenAI technologies demonstrate substantial educational potential, their implementation presents considerable equity implications that warrant systematic examination. Four key barriers to equitable adoption were identified: limited digital infrastructure and connectivity constraints, insufficient artificial intelligence literacy among stakeholders, economic barriers in terms of cost affordability and access costs, and the absence of inclusive, equity-focused institutional policies. Within the limited corpus, regional disparities were emphasised as being particularly severe, with Global South case studies identifying fundamental infrastructural constraints as primary barriers to effective technology integration. Contextual survey data from the Trust in AI Report also suggest gaps between confidence in GenAI uptake and institutional readiness in many jurisdictions, particularly in emerging economies, although these findings are not specific to the higher education sector. Findings indicate that without targeted interventions to incorporate infrastructure development, comprehensive literacy programs, access provisions, and inclusive policy frameworks, GenAI deployment has the potential to exacerbate rather than alleviate existing educational inequities in the contexts studied. This review identifies the main gaps in current research and makes



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review-derived recommendations to policymakers, higher education institutions, and practitioners to facilitate more equitable GenAI adoption in higher education environments.

Keywords Generative AI, ChatGPT, Large language models, Digital divide, Higher education, Educational technology, Equity, Inclusion, Access, Literature review, Global South

1 Introduction

The rapid proliferation of Generative Artificial Intelligence (GenAI) technologies, particularly Large Language Models (LLMs) such as OpenAI's ChatGPT, represents a paradigmatic shift with profound implications across multiple sectors, most notably within higher education [1, 2]. Following ChatGPT's public release in late 2022, adoption rates increased rapidly, with millions of users globally integrating these technologies into higher education environments for diverse applications ranging from content creation and assessment to personalised tutoring and administrative support [3–7]. The potential of GenAI to enhance pedagogical effectiveness [8, 9], enable scalable personalised learning, automate routine tasks, and augment research capacity is increasingly evident to institutions worldwide. Empirical studies report substantial growth in GenAI adoption amongst higher education stakeholders, with significant proportions of students and faculty incorporating GenAI into their academic practices [4, 10–16]. While GenAI adoption remains in its early stages, market projections suggest that the education-focused artificial intelligence sector will continue to expand in the coming years, reflecting the perceived transformative potential of GenAI technologies in the education sector.

However, such technological advancement occurs against the backdrop of persistent global inequalities [17–20]. A central manifestation of these inequalities is the multifaceted phenomenon known as the digital divide [21–26]. Earlier conceptions emphasised binary distinctions between those with and without access to Information and Communication Technologies (ICT). Contemporary scholarship, however, conceptualises the digital divide as a stratified, multi-level construct. The first-level divide concerns variation in physical access to devices, software, and the internet; the second-level divide captures variation in digital literacy and technological competencies; and the third-level divide addresses differential benefits and outcomes derived from the use of digital technologies [24–26]. Additional factors, such as affordability constraints, qualitative differences in access, infrastructural reliability, and the adequacy of the policy environment [21, 27], further shape these disparities. These differences are particularly salient between Global North and Global South contexts [21, 22], but they also manifest within nations along socio-economic, geographic, demographic, and educational lines. These studies collectively define the digital inequality context, providing a critical framework for understanding how GenAI applications may affect equity in higher education.

The convergence of GenAI and the digital divide carries significant implications for international higher education, foregrounding issues of equity, inclusion, and access. While GenAI tools hold considerable promise to democratise access to information and personalised educational support [1, 28, 29], their deployment may also exacerbate existing inequalities [29–31]. The substantial infrastructural requirements for developing and deploying sophisticated AI models, including high-capacity processing, storage, and reliable high-bandwidth connectivity, together with the costs associated with

access to powerful models, pose particular challenges in resource-constrained settings typical of many Global South nations and marginalised communities in higher-income countries [16, 30, 31]. Furthermore, embedded biases in LLMs [32–36], shaped by training data that often overrepresent Global North perspectives and linguistic contexts, risk reinforcing cultural and informational asymmetries. Global analyses of AI preparedness and policy frameworks indicate that AI capabilities and governance leadership are concentrated within a small number of economies, notably in North America and parts of Asia, widening existing capability gaps and laying the foundations for new forms of AI divide that compound the existing digital divide [37–39].

This complex interplay presents profound implications for pedagogical practice and policy development within global higher education. Institutions face dual responsibilities: realising the pedagogical and organisational benefits of GenAI while mitigating the risk of amplifying inequity. Initial institutional responses have ranged from outright prohibition due to academic integrity concerns to the development of responsible use guidelines and institutional frameworks for GenAI integration [40–43]. These responses increasingly recognise AI literacy as an essential graduate competency [1, 9, 44]. Nevertheless, institutional strategies that explicitly foreground equity and digital justice considerations have lagged behind more general pedagogical and integrity-focused responses. Without proactive interventions such as targeted infrastructure investment, comprehensive digital and AI literacy programmes, inclusive policy co-design, and context-sensitive educational innovation, the deployment of GenAI risks further excluding students with limited access or competencies, disadvantaging already marginalised communities and widening global knowledge disparities [21, 22, 29].

Despite the urgency of these challenges, empirical research that directly examines digital divide implications for GenAI adoption, attitudes and usage in higher education remains limited. Existing scholarship on GenAI in education has predominantly concentrated on pedagogical applications, institutional policy responses, and thematic analyses of opportunities and risks in relatively well-resourced contexts [8, 9, 42–47]. Studies explicitly foregrounding equity, access and infrastructural constraints in relation to GenAI are comparatively scarce, particularly in under-represented regions such as Lebanon and the broader Global South [16, 28, 30, 31, 48, 49]. Understanding these dynamics is essential for developing evidence-informed policies and practices that promote equitable access whilst harnessing GenAI's potential for inclusive educational advancement.

Consequently, the primary objective of this systematised evidence mapping review is to synthesise empirical findings published between 1 January 2023 and 30 January 2025 (including online-first articles available by that date) that examine the intersection between GenAI technologies and the digital divide within global higher education settings. Specifically, this review aims to: (1) identify key dimensions of the digital divide that function as barriers or facilitators to GenAI adoption and utilisation in higher education; (2) analyse contextual and regional variations in these challenges, with particular emphasis on insights from Global South contexts; and (3) identify critical gaps within the current research literature to inform future scholarly work and policy development. Rather than proposing that GenAI has created wholly new categories of inequality, the review interrogates how GenAI deployment overlays, reproduces, or reconfigures the established first, second, and third-level divides and emerging digital equity debates [23–26]. In this way, the contribution of the paper lies in positioning the small but rapidly

expanding post-2023 GenAI-in-education corpus within two decades of digital divide and digital equity scholarship, and in delineating where GenAI-specific equity challenges (such as differential access to subscription models, compute-intensive features, or biased outputs) appear to overlay pre-existing access, skills, and outcome gaps in higher education.

2 Methodology

A systematised scoping review (evidence mapping) methodology was employed in the study, informed by established scoping review approaches [50]. Given the nascent state of research examining the intersection of GenAI technologies and digital divide phenomena in higher education, this approach was designed to comprehensively map available evidence across diverse source types rather than conduct traditional meta-analysis. The review aimed to seek and synthesise empirical findings that explored the intersection of GenAI technologies, that is, ChatGPT and other LLMs, with digital divide patterns in higher education contexts globally.

Reporting followed PRISMA-ScR guidance, and the identification and selection process is summarised using a PRISMA 2020 flow diagram format [51, 52]. A comprehensive multi-stage search and screen strategy was employed to be methodologically rich yet adequately targeted to identify relevant literature in this rapidly evolving area of research. The overall identification, screening and inclusion process is summarised in the PRISMA flow diagram in Fig. 1.

2.1 Evidence strata

The final corpus comprised (i) primary empirical studies in higher education and (ii) secondary/contextual sources (two systematic reviews addressing broader GenAI topics [42, 43] and one foundation report [49]) from which equity-relevant insights (infrastructure, access, inclusion, policy) were thematically extracted. The systematic reviews examined ChatGPT's sectoral impacts and academic integrity, respectively, rather than the digital divide directly. All twelve sources contributed to frequency counts and thematic analysis, with systematic reviews analysed for their reported equity-related challenges and barriers rather than their primary research focus, thereby providing additional

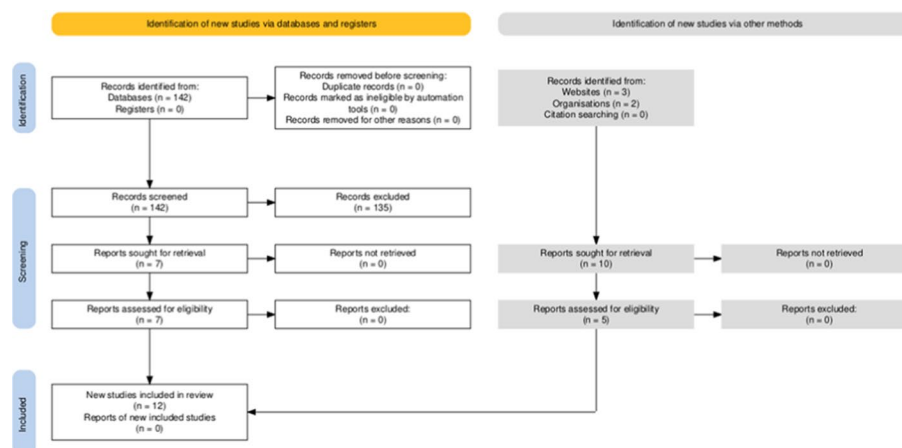


Fig. 1 PRISMA 2020 flow diagram illustrating the systematic evidence identification and selection process for mapping studies examining the intersection of Generative AI tools and digital divide phenomena in higher education

contextual insights without duplicating the underlying empirical evidence on digital divide phenomena. These descriptive counts and thematic patterns are intended only to characterise the distribution of topics within this small, purposively assembled corpus, and are not treated as statistically representative of national or regional higher education systems. The systematic reviews and the foundation report were therefore used as contextual evidence signals to sensitise the analysis to equity-related issues, rather than as additional primary datasets for any form of quantitative aggregation.

2.2 Search strategy and selection process

A comprehensive three-stage search strategy was employed to identify relevant studies examining the intersection of GenAI technologies and digital divide phenomena in higher education contexts. This approach was designed to balance methodological rigour with comprehensiveness, recognising the emerging nature of research in this domain. Given the nascent state of empirical research on the topic, the search strategy intentionally incorporated diverse evidence types, including primary studies, systematic reviews and grey literature, in order to map available evidence consistent with established practices for emerging research domains [50].

2.2.1 Stage 1: Focused database search (15–20 January 2025)

The initial stage involved a focused search within major academic databases accessed via the Central Queensland University (CQU) Library system, including Scopus, ProQuest, SpringerLink and ScienceDirect. Initial database scoping revealed significant overlap between sources, with many articles appearing in multiple databases (for example, ScienceDirect content indexed in Scopus). To avoid duplicate screening and ensure comprehensive coverage without redundancy, Scopus was selected as the primary database due to its broad multidisciplinary coverage. Supplementary searches in other databases yielded minimal additional unique results. A precise search string was constructed to capture studies explicitly linking ChatGPT, higher education, the digital divide and empirical case study methodologies:

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("ChatGPT" OR "Generative AI" OR "Large Language Model*")  
AND ("higher education" OR university* OR college*) AND  
("digital divide") AND ("case study")
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Filters were applied to limit results to peer-reviewed journal articles published in English between 1 January 2023 and the search date. Results were initially sorted by relevance or publication date (newest first). This highly targeted search yielded 142 initial results from which titles and abstracts were screened against strict inclusion criteria (Table 1), identifying only two articles that met all strict criteria. The focused database search was completed on 20 January 2025. Search results were exported with full bibliographic details and imported into reference management software for systematic screening. This limited output reflects the genuinely emerging state of empirical research directly addressing this specific intersection using a case study approach.

2.2.2 Stage 2: Expanded database search (21–25 January 2025)

Recognising the limitations of the narrow initial search and the likelihood that relevant studies might employ broader terminology for equity issues, the same 142 initial results

Table 1 Summary of inclusion and exclusion criteria

Criterion	Inclusion condition	Exclusion condition
Publication type	Peer-reviewed empirical studies (quantitative, qualitative, mixed-methods), case studies; empirical grey literature reporting primary data; systematic reviews used as contextual secondary sources (not primary empirical evidence)	Narrative reviews; systematic reviews treated as primary empirical evidence; editorials/opinion pieces; conceptual/technical papers without empirical data; book chapters
Timeframe	Published between 1 January 2023 and 30 January 2025 (including online-first articles available by that date)	Published before 2023
Language	English	Non-English
Technology focus	Explicitly mentions ChatGPT, Generative AI or LLMs	General AI/EdTech without specific reference to GenAI/LLMs
Educational context	Higher education (university/college students, educators, institutions)	Primarily K–12, corporate training or non-educational settings
Equity/access focus	Addresses dimensions of digital divide, access, inclusion, equity, infrastructure or literacy	No mention of digital equity, access issues or related barriers
Methodology	Empirical studies (quantitative, qualitative, mixed-methods), case studies, reports; systematic reviews used for contextual thematic extraction only	Non-empirical, purely theoretical/technical papers; narrative/opinion-only syntheses

were re-screened using expanded inclusion criteria. The criteria related to the “digital divide” were broadened conceptually to include studies that, whilst potentially not using the exact phrase “digital divide”, addressed core components such as digital access, inclusion, equity, infrastructure challenges or digital literacy gaps in the context of GenAI adoption in higher education. Additional database searches were conducted using expanded search terms:

("ChatGPT" OR "Generative AI" OR "Large Language Model")
AND ("higher education" OR university* OR college*) AND
("access" OR "inclusion" OR "infrastructure" OR "equity"
OR "digital literacy")*

The requirement for a “case study” methodology was also relaxed to include other empirical approaches (for example, surveys, mixed-methods) or reviews focusing on the relevant themes. This expanded screening process identified an additional five relevant papers from the original corpus and supplementary searches. This stage included two systematic reviews [42, 43], which, although not primary empirical studies, provided valuable thematic synthesis on digital access, infrastructure and equity issues related to GenAI adoption. Their inclusion was justified as these reviews synthesised relevant empirical evidence and identified key themes directly pertinent to the research question, following established practices for evidence mapping in emerging domains [50]. This brought the total number of included studies after the second stage to seven.

It should be noted that the systematic reviews by Bittle and El-Gayar [42] and Hussein et al. [43] were included during this stage not as primary empirical evidence but for targeted thematic extraction of equity-related insights relevant to the research question. These reviews underwent secondary analysis to identify themes related to infrastructure barriers, access challenges and inclusion issues rather than synthesis of their primary study findings, thereby avoiding double-counting of empirical evidence while capturing relevant theoretical and thematic insights from broader GenAI research.

2.2.3 Stage 3: Grey literature and manual search (26–30 January 2025)

To further enhance comprehensiveness and capture research potentially not indexed in traditional academic databases or published rapidly, the final stage involved systematic manual searching of alternative sources. Targeted searches were conducted on Google Scholar, ResearchGate and relevant institutional or organisational repositories known for publishing reports on education and technology (for example, arXiv, the Abdulla Al Ghurair Foundation, university institutional repositories). The same expanded search terms employed in stage 2 were utilised, with systematic screening of the first 100 results from each source.

Inclusion criteria remained consistent, focusing on empirical data or substantive reports related to GenAI/ChatGPT adoption in higher education that addressed dimensions of digital access, equity or infrastructure. This stage was particularly important for identifying context-specific reports and studies from the Global South. This supplementary search provided five additional relevant sources, including case studies and foundation reports [16, 28, 30, 31, 49], bringing the final number of studies included for analysis to twelve.

2.3 Inclusion and exclusion criteria

A summary of the key inclusion and exclusion criteria applied across the stages is presented in Table 1. These criteria were consistently applied across all three search stages, with stage 2 representing an expansion of terminology rather than fundamental changes to study requirements.

2.4 Data extraction and analysis

Data from the final twelve included studies were extracted systematically. Key information recorded included author(s), publication year, title, publication source/type, study methodology, geographical focus/context and specific findings related to GenAI/ChatGPT adoption and the digital divide (including reported barriers, facilitators, impacts and recommendations). A standardised data extraction form was developed, encompassing study characteristics (author, year, methodology, geographic context), technology focus (specific GenAI tools examined), equity dimensions addressed and key findings related to digital divide barriers and facilitators. Title and abstract screening was conducted by the lead reviewer, with full-text assessment decisions reviewed by the research team to ensure consistency with inclusion criteria. Data extraction was conducted systematically by R.J., with extracted data verified through discussion amongst the research team to ensure consistency and completeness.

A qualitative thematic synthesis approach was employed to analyse the extracted data, drawing on established frameworks for systematic review synthesis and digital divide theory [24–26, 50]. Findings across the studies were coded and categorised based on recurring dimensions of the digital divide (for example, infrastructure barriers, digital literacy gaps, cost/affordability issues, policy responses, regional disparities). Initial coding was conducted deductively using established digital divide dimensions, supplemented by inductive identification of emerging themes specific to GenAI adoption contexts.

The coding framework employed a hybrid deductive–inductive approach, beginning with van Dijk’s three-level digital divide framework (access barriers, skills gaps, usage

outcomes) as initial organising categories [24, 25]. Additional codes emerged inductively during analysis of study findings. For example, the “infrastructure barriers” category encompassed codes such as “electricity shortages” (derived from Boustani et al. [30]: “electricity shortages ... severely limited students’ ability”), “connectivity issues” (from Choi et al. [31]: “fundamental infrastructure deficits”) and “device inadequacy”. The “digital literacy gaps” category included codes like “faculty unpreparedness” (from Egin et al. [53]: “low AI literacy among both students and educators”) and “student overconfidence” (informed by Trust in AI survey data showing confidence–support gaps [39]). The Trust in AI 2025 survey was treated solely as background contextual literature that helped sensitise the construction of the coding frame and was not incorporated as an additional empirical dataset to be triangulated with the mapped GenAI studies. Initial coding was conducted by R.J., with regular supervisory meetings held to review coding decisions, discuss emerging themes and ensure consistency in theme assignment. Coding discrepancies and ambiguous cases were resolved through collaborative discussion until consensus was reached among the research team.

This thematic analysis aimed to identify dominant patterns, contrasts and gaps in the literature concerning the interplay between GenAI and educational equity in higher education. The relatively modest corpus reflects the genuine nascency of empirical research specifically examining GenAI–digital divide intersections in higher education, reinforcing both the timeliness of this evidence mapping exercise and the critical need for expanded research in this domain. In line with this evidence-mapping orientation, the synthesis is explicitly descriptive and configurative rather than aggregative: no pooled effect sizes were calculated, and no claims are made about the prevalence or magnitude of impacts beyond the specific institutional and national contexts represented in the included studies.

2.5 Quality assessment

Study quality was assessed using criteria adapted from established frameworks, including the Critical Appraisal Skills Programme (CASP) checklists and the Joanna Briggs Institute (JBI) tools, modified to accommodate the mixed methodologies encountered. Assessment focused on four key dimensions: methodological clarity, sample appropriateness, data collection rigour and relevance to the research objectives. Quality assessment was conducted by R.J., with review and discussion of assessments by the research team to ensure consistency.

For empirical studies, assessment criteria included the clarity of the research design, the appropriateness of the methodology to the research questions, an adequate description of the sample, transparent data collection procedures and logical analysis approaches. For systematic reviews, the assessment focused on the comprehensiveness of the search strategy, the appropriateness of the synthesis methods and the transparency of reporting. For grey literature sources, evaluation emphasised empirical content quality, methodological transparency where reported, institutional credibility and clear distinction from purely advocacy materials.

Given the nascent state of research in this domain and the inclusion of grey literature for comprehensiveness, studies were not excluded based on quality scores. Findings from higher-quality studies were given greater emphasis in the synthesis, particularly when conflicting evidence emerged. Foundation reports and institutional surveys were

included based on their empirical content and relevance to policy implementation, whilst maintaining focus on evidence-based rather than purely advocacy materials. Quality limitations were explicitly noted in the discussion of individual study findings and considered when drawing overall conclusions. Accordingly, when subsequent sections refer to some themes as being more or less strongly evidenced than others, this should be understood as indicating their relative support within this small, heterogeneous corpus rather than implying any form of statistical generalisability to wider populations.

2.6 Methodological limitations and justifications

Several methodological adaptations were necessitated by the emerging nature of the research domain. The evidence-mapping approach employed represents a pragmatic response to the limited empirical research base specifically addressing GenAI–digital divide intersections in higher education [50]. The temporal constraint to 1 January 2023 through 30 January 2025 reflects the genuine emergence of generative AI technologies in educational contexts rather than an arbitrary restriction, as widespread adoption of tools such as ChatGPT commenced only in 2023 following its November 2022 release. As an evidence map of a nascent field, the corpus includes diverse evidence types. Two systematic reviews addressing broader GenAI topics were included for thematic extraction of equity-relevant insights rather than analysis of their primary research focus [42, 43], providing additional contextual perspectives while maintaining focus on digital divide phenomena specifically. A further limitation is that the empirical studies included in the map are predominantly institution- or programme-specific investigations that rely on relatively modest, often convenience-based samples (for example, a single private university in Lebanon in Boustani et al., a cohort of 195 engineering students in Elkefi et al., and case studies in Vietnam and online course settings in Nguyen et al. and Lin et al.) [6, 16, 30, 54]. The review therefore does not treat these studies as representative of entire national or regional higher education systems; instead, they are interpreted as analytically rich exemplars that signal how GenAI-related equity issues are beginning to manifest within particular configurations of the digital divide.

The inclusion of diverse evidence types, including systematic reviews for thematic extraction and grey literature sources, was deemed necessary to comprehensively capture available insights in this rapidly evolving field [50]. This approach aligns with established guidance for evidence synthesis in emerging technology domains where traditional systematic review approaches may yield insufficient primary studies for meaningful synthesis. The decision to broaden inclusion criteria across the three search stages was made transparently and systematically rather than post hoc, recognising the exploratory nature of evidence mapping in nascent research areas.

3 Related work

The systematic investigation of educational applications of generative AI technologies has been carried out under various methodological frameworks, with increasing emphasis on issues of equity and technology access. A comprehensive analysis of existing reviews uncovers emergent research streams while also precipitating urgent gaps in empirical understanding of technology-enabled educational equity. In parallel, a substantial pre-GenAI literature on the digital divide and digital inequality in education

has developed over more than two decades, providing the access–skills–outcomes and digital equity frameworks against which contemporary GenAI developments in higher education must be interpreted [23–26].

3.1 Foundational systematic reviews in AI education research

A comprehensive analytical framework was established through a systematic review of 146 articles published between 2007 and 2018 in which the use of artificial intelligence in higher education was classified into four broad categories: profiling and prediction, intelligent tutoring systems, assessment and evaluation, and adaptive systems and personalisation [45]. It was observed that disciplinary authorship patterns demonstrated significant bias, with merely 8.9% of first authors affiliated with education departments, whilst computer science and STEM disciplines dominated at 62%. Most significantly, the analysis revealed what was characterised as a “stunning result” regarding the absence of critical reflection on pedagogical and ethical implications, with only 1.4% of studies addressing equity considerations or digital access challenges [45]. Crucially, this review predates the emergence of LLM-based GenAI tools and does not explicitly mobilise the now-canonical multi-level digital divide frameworks, thereby leaving the relationship between AI applications and differentiated access, skills and outcomes largely implicit [23–25].

This foundational work was subsequently extended through the first meta-systematic review in the domain, wherein 66 evidence syntheses published between 2018 and 2023 were analysed using tertiary review methodology [44]. It was demonstrated that whilst adaptive systems and personalisation constituted the most frequently examined applications, systematic deficiencies persisted across ethical considerations, methodological rigour, and international collaboration frameworks. The analysis determined that 40.9% of reviews conveyed explicit demands for more ethical considerations, thus creating a “call for increased ethics, collaboration, and rigour” in the research community [44]. From a digital equity perspective, this meta-review confirms that equity and access are recognised as concerns in the AI-in-education literature but are usually treated as subsidiary themes rather than as primary organising constructs, and its temporal scope ends just as post-ChatGPT empirical work begins to emerge.

3.2 Systematic examination of generative AI and digital equity

The emergence of generative AI and large language models has necessitated more targeted analyses of equity implications. A systematic review of 41 studies published between 2021 and 2024 examined generative AI impacts on academic integrity in higher education environments [42]. Although academic integrity concerns were the primary analytic focus, broader ethical implications were identified, including bias perpetuation risks, potential exacerbation of the digital divide, and environmental sustainability concerns associated with AI technology deployment [42]. However, digital divide issues in this review are largely flagged as emerging risks or agenda items rather than being operationalised through specific access, skills, or outcome indicators grounded in the digital divide literature.

Complementary analysis was carried out through a systematic review of generative AI trends and learning applications, wherein 37 empirical studies were examined to determine the comparative effectiveness of AI-generated versus human-provided feedback

mechanisms [46]. It was reported that AI-generated feedback did not consistently produce superior linguistic progression outcomes compared to traditional human tutoring approaches, and that its benefits were contingent on careful pedagogical integration [46]. The review also identified confidence–support gaps, particularly evident in emerging economies, where student confidence in AI tools often exceeded institutional readiness and faculty training provisions [46]. Yet, as with the academic-integrity-focused synthesis, equity and digital divide considerations surface as important contextual observations rather than as the central analytic lens, and the primary emphasis remains on learning effectiveness and classroom integration.

3.3 Digital divide considerations in large language model applications

Recent systematic reviews have begun to address digital equity implications more explicitly. A survey-style systematic review of large language models in education highlighted “insufficient educational resources” and “equity concerns” as intrinsic implementation problems, noting that access barriers arise when some families cannot afford AI learning tools or when students lack access to the necessary technological facilities for tools like ChatGPT [9]. The analysis emphasised that educational equity constitutes “the cornerstone of social development” and argued that systematic interventions are required to address digital divide challenges effectively in the context of LLM deployment [9].

Moreover, a systematic review of large language models in personalised learning environments, synthesising 55 publications from 2020 to 2024, identified equity in learning access as one of the most crucial ethical concerns [8]. It was argued that AI-based tools should not disproportionately benefit students who are already better positioned with regard to access to technology while harming marginalised and underprivileged communities [8]. The analysis emphasised that the digital divide must be mitigated so that all learners, regardless of their socioeconomic status, are offered access to AI-based learning resources on an equal footing. Notwithstanding these important observations, both reviews primarily conceptualise equity at a high level and do not systematically map GenAI-related inequalities onto the established first-, second- and third-level digital divides or the broader digital equity interventions synthesised in prior work [23–26].

3.4 Methodological limitations and geographical bias

A further systematic literature review of empirical generative AI studies in education depicts persistent methodological constraints that limit insight into digital equity dynamics. A review of 28 empirical articles on generative AI technologies drawn predominantly from higher-impact journals reported limited geographical diversity and a lack of focus on resource-poor educational environments [47]. The predominance of research carried out in technologically advanced environments has resulted in an incomplete picture of how digital divide dimensions influence GenAI adoption in diverse learning environments across the globe [47]. Specifically, studies from regions characterised by infrastructural instability, constrained connectivity, or pronounced affordability barriers remain under-represented, despite these contexts being central to long-standing digital divide debates [20–22, 27].

In addition, an institutional policy analysis across 116 US Research-1 institutions revealed that although 63% encouraged the deployment of GenAI, the guidance frameworks largely emphasised writing applications with minimal regard for STEM disciplines

or comprehensive equity considerations [40]. This policy–practice gap suggests that institutional responses are not yet systematically aligned with the complex access and literacy issues highlighted in the broader digital divide and digital equity scholarship [23, 26]. Taken together, the existing GenAI-focused reviews and policy analyses therefore provide only partial visibility of how structural access, skills, and outcome inequalities intersect with GenAI adoption in higher education.

3.5 Research gaps and analytical positioning

Against this backdrop, the analysis of the literature reveals several specific gaps that motivate the present evidence-mapping review. First, to the best of the authors' knowledge, no prior review has taken the intersection of GenAI technologies and the multi-level digital divide as its primary and explicit focus of investigation in higher education, despite digital inequality frameworks being well established in adjacent work [23–26]. Existing GenAI syntheses acknowledge equity issues but typically subsume them under broader ethical, pedagogical, or academic integrity themes. Second, geographical bias in favour of technologically advanced settings has constrained understanding of Global South perspectives and implementation challenges, particularly in countries and institutions grappling with unreliable infrastructure, high connectivity costs, or constrained device access [16, 21, 22, 30, 31]. Third, the temporal limitations of existing reviews mean that much of the empirical evidence synthesised predates the period of mass GenAI diffusion following the public release of ChatGPT in late 2022, thereby limiting insight into current implementation dynamics in higher education [8, 9, 42, 44, 47].

The present systematised evidence-mapping review seeks to address these gaps by synthesising in-depth empirical studies that expressly examine GenAI tools and digital divide intersections in global higher education contexts. Specifically, the review extends digital divide scholarship in two ways. First, it examines how GenAI's distinctive socio-technical characteristics create novel configurations of the established three-level divide. Unlike earlier educational technologies, GenAI exhibits: (a) subscription-based access tiers that introduce new economic stratification between free and premium users within the same institutional cohort; (b) compute-intensive features (such as image generation, extended context windows, or real-time web integration) that require high-bandwidth connectivity and premium subscriptions, thereby compounding first-level access barriers; and (c) embedded linguistic and cultural biases in training data that impose culturally-specific literacy requirements, extending second-level skill gaps beyond generic digital competencies to encompass critical evaluation of algorithmic outputs shaped by Global North data dominance.

Second, the temporal focus on 1 January 2023 to 30 January 2025 captures a critical transition period characterised by a structural mismatch between bottom-up GenAI diffusion and institutional preparedness. The mass availability of ChatGPT from November 2022 created unprecedented conditions wherein students adopted general-purpose LLMs at scale ahead of curriculum integration, faculty development, or policy formulation—a pattern fundamentally different from earlier educational technology adoptions that were typically institution-led or mediated through learning management systems. This temporal specificity enables analysis of how rapid, student-driven adoption intersects with pre-existing digital divides in ways that were not observable in prior EdTech deployments.

The review is configurative rather than aggregative in intent: it does not aim to estimate population-level effects or prevalence, but to map how GenAI-related equity issues are beginning to materialise within specific institutional and national settings characterised by different configurations of the digital divide. Moreover, the explicit emphasis on Global South contexts and resource-constrained settings seeks to move beyond the geographical bias of the previous literature. In the discussion, the patterns identified in the mapped studies are situated alongside large-scale survey evidence on trust, attitudes and use of AI across 47 countries [39]; this external dataset is treated solely as background contextual literature that helps to interpret and problematise emerging themes, rather than as an additional empirical source to be triangulated or statistically combined with the reviewed studies.

Therefore, this review contributes to knowledge development by providing a GenAI-specific evidence map anchored in established digital divide and digital equity scholarship, offering empirically grounded directions for research advancement and practical implementation strategies to support more equitable GenAI adoption in global higher education settings. Complementing these equity frameworks, Rogers' Diffusion of Innovations theory [55] provides an interpretive lens for understanding technology adoption patterns in higher education. Rogers identifies five key attributes that influence adoption rates: relative advantage (perceived benefit over existing practices), compatibility (alignment with adopter values and needs), complexity (ease of use), trialability (opportunity to experiment), and observability (visibility of results). This theoretical lens is used to interpret the emerging 'adoption paradox' observed in the GenAI literature, where high student uptake persists despite structural constraints. The framework helps explain how perceived relative advantage and compatibility can drive rapid bottom-up diffusion even when institutional scaffolding, formal training, and policy frameworks are absent or underdeveloped, creating the efficacy–support asymmetry documented in both the reviewed studies and broader survey evidence.

4 Results and analysis

This section presents the results of the twelve included studies synthesised within this review. Firstly, an overview of the characteristics of the included studies is provided. Subsequently, the key findings on the main dimensions of the digital divide interacting with the adoption and use of GenAI, primarily ChatGPT, in higher education environments are presented. Throughout this section, the terms “evidence”, “coverage” and “themes” are used in a configurative and descriptive sense, referring to patterns across the twelve included publications rather than to statistically representative or generalisable claims about countries or regions.

4.1 Overview of included studies

The final selection comprised twelve publications: eight empirical journal articles, two systematic review journal articles, one arXiv preprint and one foundation report. The studies employed a range of methodologies, including case studies, mixed-methods designs, quantitative and qualitative empirical studies, topic modelling, systematic reviews and policy-oriented reports. Geographically, the studies varied: several had an unspecified or global focus, while others centred on specific regions or countries, including Lebanon, developing countries more broadly, Southeast Asia, Sierra Leone, and a

range of higher education contexts located in comparatively better-resourced systems. The equity focus also varied across multiple dimensions, encompassing access, inclusion, infrastructure, digital literacy and policy implications. Table 2 provides a summary of the included studies.

Most empirical studies employed single-institution or small-sample designs (for example, 195 engineering students in the multi-country mixed-methods study by Elkefi et al. [16]), or focused on a limited set of institutions or communities. The synthesis therefore traces converging patterns across these specific sites, without claiming statistical representativeness for any country or region. Case-study designs account for five of the twelve studies, while only one purely quantitative survey and one mixed-methods study were identified, underscoring the exploratory and context-bound character of the current evidence base.

Table 2 Characteristics of included studies (n = 12)

References	Title	Publication type	Methodology	Primary equity focus
[4]	Students' use of large language models in engineering education: A case study on technology acceptance, perceptions, efficacy, and detection chances	Journal article	Case study	Access
[6]	Artificial Intelligence (AI) in Education: A Case Study on ChatGPT's Influence on Student Learning Behaviors	Journal article	Case study	Access
[53]	Let's Talk About EdTech! A Topic Modelling Analysis of AI Tools and Pre-Service Teachers' Perspectives	Journal article	Topic modelling	Inclusion
[56]	Teachers Can Use It with Their Littles: Using Case Studies to Explore Preservice Teachers' Perceptions of Technology	Journal article	Case study	Inclusion
[43]	ChatGPT's Impact Across Sectors: A Systematic Review of Key Themes and Challenges	Journal article	Systematic review*	Infrastructure
[28]	Writing with ChatGPT in a Context of Educational Inequality and Digital Divide	Journal article	Qualitative case study	Digital literacy
[42]	Generative AI and Academic Integrity in Higher Education: A Systematic Review and Research Agenda	Journal article	Systematic review*	Inclusion
[30]	Leveraging ICT and Generative AI in Higher Education for Sustainable Development: The Case of a Lebanese Private University	Journal article	Quantitative survey	Infrastructure
[49]	IGNITE TechTalk Lebanon: Empowering Youth in Lebanon through Digital Upskilling and Jobs	Foundation report	Report (mixed empirical content)	Digital literacy
[16]	Use of ChatGPT for education by engineering students in developing countries: a mixed-methods study	Journal article	Mixed methods	Access
[54]	A Case Study Investigating the Utilization of ChatGPT in Online Discussions	Journal article	Case study	Digital literacy
[31]	Are LLMs Useful in the Poorest Schools? The-Teacher. AI in Sierra Leone	arXiv preprint	Case study / feasibility study	Infrastructure

Systematic reviews [42, 43] were included for contextual thematic synthesis relevant to digital access and equity in GenAI adoption rather than as primary empirical evidence. The "Primary equity focus" column indicates the digital divide dimension most explicitly discussed in each publication; in several studies, equity issues arise mainly in the discussion or recommendations rather than as the central research question, and in this review such instances are treated as secondary, contextual evidence about digital divide dynamics

4.2 Thematic findings: GenAI and the digital divide in higher education

The analysis revealed five interrelated themes concerning the challenges and opportunities presented by GenAI adoption in the context of the digital divide, as shown in Fig. 2. The remainder of this section groups these themes into three primary clusters. Where counts such as “7/12 studies” are reported, they refer simply to how many of the twelve included publications contained that theme as a salient topic in the analysis, not to the strength of causal evidence or to population prevalence.

4.2.1 Structural barriers to equitable access (frequently reported: 7/12 studies)

Infrastructure inadequacies and economic constraints emerged as fundamental barriers hindering equitable GenAI adoption. Studies focusing on developing countries and Global South contexts highlighted significant challenges, with Elkefi et al. [16] emphasising that whilst adoption rates among engineering students who had access were high, the lack of robust infrastructure remained a major concern for full participation. Boustani et al. [30] provided a detailed account of the Lebanese context, where electricity shortages, poor internet connectivity and the high cost of mobile data severely limited students’ ability to access and utilise ICT and GenAI tools effectively. Similarly, Choi et al. [31] explored the feasibility of LLMs in resource-poor schools in Sierra Leone, implicitly pointing to fundamental infrastructure deficits that constrain any realistic deployment of GenAI-based interventions.

Economic barriers compound these infrastructure challenges, with direct and indirect costs associated with accessing and utilising GenAI tools constituting significant obstacles. Boustani et al. [30] explicitly linked the high cost of mobile data and potentially premium AI tools to limited access among Lebanese students. Whilst many basic GenAI tools are initially free, the computational power required, potential costs of advanced versions or specialised platforms, and the underlying cost of devices and reliable internet access create economic hurdles, particularly in low-income contexts. This economic dimension intersects closely with infrastructure challenges, collectively limiting

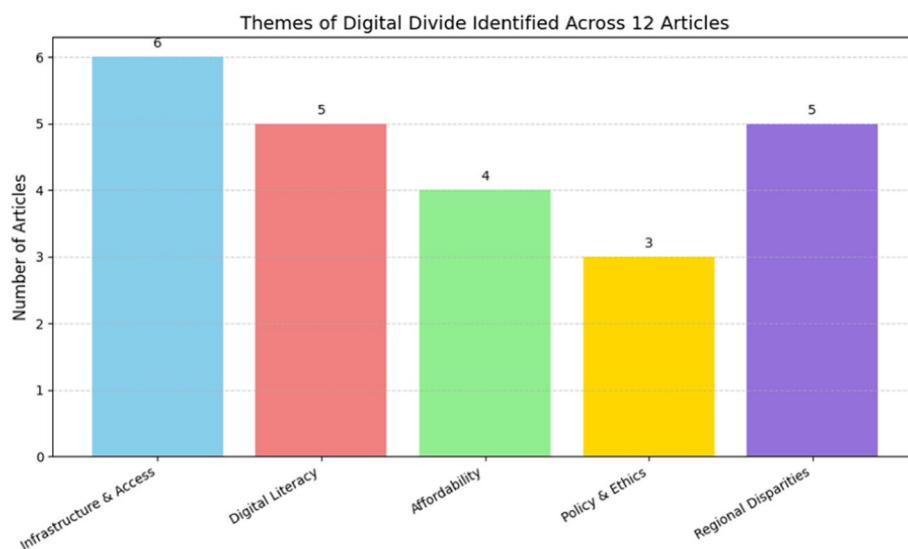


Fig. 2 Digital divide themes, with infrastructure barriers coded as the primary constraint in 7 of the 12 reviewed studies. The figure summarises only the thematic salience of barriers within the twelve included publications and should not be interpreted as a representation of national or regional conditions

equitable participation. The systematic review by Hussein et al. [43] also identified infrastructure as a recurring challenge impacting ChatGPT's broader adoption across sectors, which has clear implications for education. Taken together, these studies indicate that access and affordability constraints are repeatedly foregrounded as central barriers in the early empirical literature, but the evidence remains tightly bound to the specific institutions, programmes and communities investigated.

4.2.2 Capacity and governance challenges (reported in 5/12 studies)

Beyond physical access, significant capacity gaps and inadequate institutional responses emerged as critical barriers to equitable GenAI integration. Digital literacy deficits and the skills required to use GenAI tools effectively and ethically represented a substantial second-level divide. Eğin et al. [53], analysing pre-service teachers' perspectives using topic modelling, found generally low and uneven AI literacy among both students and educators, warning that this could exacerbate inequities if not addressed through capacity-building. George and Mattathil [48] and Santiago-Ruiz [28] similarly reported limited AI literacy among students and educators in specific higher education settings, with reliance on ChatGPT for information retrieval and limited prompt-design or critical-evaluation proficiency, underscoring the need for targeted upskilling in higher education. Lin et al. [54] observed diversity in student AI literacy and technical proficiency in online discussions, suggesting that disparities in AI knowledge could affect participation and recommending further research into non-use, implicitly linking non-engagement to literacy and confidence barriers.

In several of these studies, equity-related concerns arise primarily in the implications and recommendations sections rather than as the central research question. For example, Lin et al. [54] and Beisly and Abeyrathna Herath Mudiyanseleg [56] focus analytically on participation, technology use and pedagogical design, but explicitly caution that uneven access and literacy may produce differential benefits for learners. In this review, such contributions are therefore interpreted as providing indicative, secondary evidence about digital divide dimensions linked to GenAI, rather than as direct evaluations of equity interventions.

Institutional and policy responses were found to be inadequate or potentially counterproductive. The review by Bittle and El-Gayar [42] highlighted concerns regarding institutional policies, academic integrity and ethical considerations surrounding GenAI use, and noted that many institutional responses have yet to engage substantively with equity issues. Other studies observed that unequal access raises ethical questions, as students with access may gain advantages over those without comparable infrastructure or literacy [16, 28, 30]. Furthermore, institutional or national bans on GenAI tools, often implemented due to integrity concerns, were cautioned against, as they risk widening digital and skill gaps by creating unequal learning opportunities. Several studies called for proactive policy interventions, including government investment in infrastructure, the development of ethical guidelines, inclusion of AI ethics and literacy in curricula, and institutional strategies promoting equitable use. The Abdulla Al Ghurair Foundation report [49] strongly advocated for digital upskilling programmes in Lebanon, highlighting the need to equip youth with relevant competencies for the AI era in tandem with infrastructure and affordability measures.

4.2.3 Implementation dynamics and contextual variations (reported in 6/12 studies)

Despite documented barriers, the analysis revealed complex adoption patterns characterised by significant interest and contextual variations in GenAI implementation. Several studies indicated substantial interest in and adoption of GenAI tools among higher education stakeholders, where access permitted. Elkefi et al. [16] reported high adoption rates among engineering students in developing countries, conditional on availability of devices and connectivity. Bernabei et al. [4] explored technology acceptance and perceptions among engineering students, while Nguyen et al. [6] investigated ChatGPT's influence on student learning behaviours in a Southeast Asian higher education context. Teacher perceptions were examined in Egin et al. [53] and Beisly and Abeyrathna Herath Mudiyanseleg [56], revealing both perceived pedagogical potential and concerns regarding classroom integration and equity. This constellation of findings suggests a strong pull towards utilising GenAI tools wherever they are available, thereby making the equity and access issues described above even more critical to address.

The findings also highlighted the importance of contextual considerations in understanding GenAI adoption patterns. Specific challenges were prominent in certain regions, with the Lebanese context characterised by severe infrastructure constraints compounded by economic factors [30, 49]. Studies focusing on developing countries emphasised widespread access issues despite high interest and positive attitudes towards GenAI [4, 6, 16]. The conditions in extremely resource-constrained environments, such as the schools studied in Sierra Leone, presented fundamental feasibility questions concerning whether and how GenAI might realistically be introduced [31]. By contrast, studies conducted in comparatively better-resourced higher education systems documented challenges that were less about physical access and more about uneven preparedness, varying levels of AI literacy and uncertainty about appropriate pedagogical integration [53, 56]. This variation emphasises that whilst broad themes recur across settings, solutions and interventions must be context-sensitive, recognising the diverse challenges and opportunities present across different educational environments.

Figure 3 visualises the regional distribution of barrier themes as they appear in the twelve included publications. The barrier intensity levels (0–3: not mentioned, mild, moderate, severe) were assigned based on explicit decision rules applied consistently across studies: (0) no mention of the barrier; (1) brief acknowledgment or minor concern noted; (2) substantial discussion with supporting qualitative or quantitative evidence but where the barrier is not the primary focus; (3) central barrier with detailed documentation and/or quantitative evidence. Coding was conducted at the level of each study's immediate empirical context (for example, a given university, programme or national system), and then aggregated descriptively for the figure. The shading in Fig. 3 should therefore be interpreted only as a visual summary of the salience of particular barriers in the twelve reviewed studies, rather than as a statistical characterisation of whole regions or countries.

These themes demonstrate uneven but converging coverage across the twelve publications. Infrastructure and literacy challenges are the most consistently discussed barriers, whereas policy interventions and longer-term educational impacts receive comparatively limited attention. Given the small number of studies and their context-specific designs, the patterns identified here should be read as early signals that warrant further

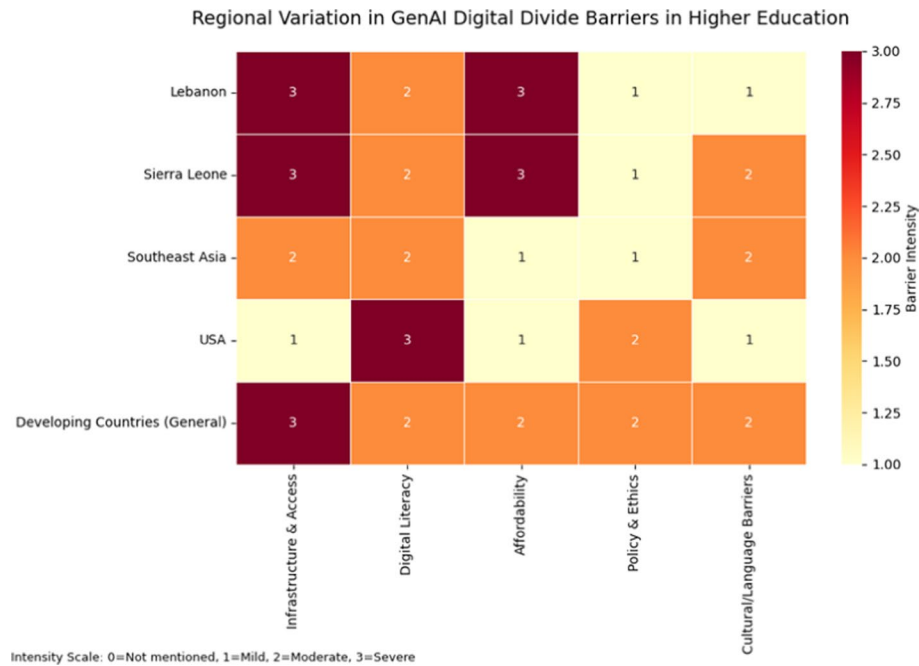


Fig. 3 Regional variation in GenAI-related digital divide barriers across reviewed studies, based on qualitative barrier-intensity coding of the twelve included publications. Shading reflects barrier salience within the specific studies reviewed and should not be interpreted as representative of entire national or regional higher education systems

Table 3 Research maturity and evidence gaps analysis

Research domain	Empirical coverage	Methodological strength	Geographic representation	Critical gap identified
Infrastructure barriers	High (7/12, 58%)	Moderate to strong (multiple case studies and one survey with clear reporting)	Predominantly Global South-focused (5/7 studies)	Limited evaluation of concrete intervention strategies and policy responses
Digital literacy	Moderate (5/12, 42%)	Mixed (case studies and qualitative analyses, limited comparative work)	Diverse contexts (emerging and better-resourced systems)	Absence of validated AI-literacy competency frameworks linked to GenAI use in higher education
Policy interventions	Low (3/12, 25%)	Generally weak (mostly descriptive or secondary in focus)	Limited geographic scope (few systems examined in depth)	Lack of empirical studies evaluating policy designs or institutional implementation from an equity perspective
Longitudinal impacts	None (0/12, 0%)	Not assessed	No longitudinal coverage identified	Major methodological gap in understanding longer-term effects of GenAI on digital inequality and educational outcomes
Economic barriers	Moderate (5/12, 42%)	Surface-level analysis (costs noted but rarely modelled)	Context-specific (notably Lebanon and selected developing-country settings)	Lack of cost–benefit analyses and affordability studies examining paid versus free GenAI access for students

investigation rather than as definitive or generalisable estimates of how GenAI and the digital divide interact across higher education systems.

Table 3 characterises research maturity within the twelve-study corpus across four heuristic dimensions. *Empirical coverage* reflects theme frequency, classified qualitatively as High (theme central to $\geq 50\%$ of studies, i.e., $\geq 6/12$), Moderate (25–49%, i.e., 3–5/12), Low ($< 25\%$, i.e., 1–2/12), or None (0/12). *Methodological strength* represents a

corpus-level qualitative assessment based on study design clarity (explicit reporting of sampling, data collection, and analysis procedures), transparency (availability of methodological detail enabling replication assessment), and coherence (alignment between stated research questions and methods employed), rather than formal risk-of-bias scoring. *Geographic representation* indicates the predominant settings of studies addressing each theme and does not imply statistical representativeness for those regions. *Critical gap identified* is an interpretive synthesis label summarising where current evidence is thinnest or least developed, derived from analysis of what studies explicitly identify as missing or what logical extensions of current findings are absent from the corpus. These categories map research maturity and identify priorities for future work, not quantitative indicators of prevalence or effect sizes.

Overall, the findings indicate that while immediate barriers to GenAI access and use, particularly infrastructure constraints, affordability and uneven digital literacy, are already being documented across diverse settings, the mechanisms for addressing these challenges and their long-term implications for educational equity remain poorly understood. Rather than supporting broad extrapolation to whole regions or sectors, the current evidence base is best interpreted as an early, patchwork map of how GenAI is intersecting with pre-existing digital divides in specific higher education contexts. This underscores the need for carefully designed comparative and longitudinal studies, accompanied by robust evaluation of policy and pedagogical interventions, if GenAI is to contribute meaningfully to digital equity goals rather than entrenching existing inequalities.

5 Global survey context: trust in AI report 2025

During the final stages of this review, the Trust in AI Report 2025 became available, providing a rich source of cross-national survey data on AI adoption, confidence and governance. While this report was not part of the systematised search corpus, it offers high-quality, independently collected evidence on public and student AI use across multiple countries. In this section, the report is treated solely as supplementary background literature that contextualises the twelve-study evidence map, rather than as an additional data source for formal triangulation or statistical validation of the review findings [39]. No indicators from the Trust in AI Report were used in study selection, coding, or the construction of Figs. 1, 2, and 3; instead, selected descriptive results are cited to illustrate whether the qualitative patterns observed in the higher-education corpus resonate with broader global trends.

The Trust in AI study surveyed more than 48,000 respondents across 47 countries using nationally representative research panels between November 2024 and January 2025, with instruments administered in local languages using professional translation processes [39]. Because the student module is not designed to resolve within-country educational subgroups or institutional types, no country-level or population-type comparisons are attempted here; results are mobilised only at the level of aggregate patterns that speak to themes already identified in the GenAI–digital divide literature.

5.1 Trust in AI report methodology and metric selection

The Trust in AI Report employed online surveys with a minimum of 1 000 respondents per country, collecting data on AI usage patterns, trust levels, training exposure, and

perceptions of organisational governance. A dedicated student module examined self-reported AI use in study, perceived knowledge and efficacy, and the availability of institutional support and policies [39]. For contextual comparison with the present review, attention is restricted to three sets of indicators that correspond directly to themes identified in Sects. 3 and 4: (i) student confidence and use of AI (relating to adoption patterns); (ii) exposure to AI-related education and training (relating to capacity gaps and digital literacy); and (iii) perceived adequacy of institutional safeguards and policies (relating to governance and policy responses). These indicators are interpreted descriptively and qualitatively, without re-weighting or integrating them into the evidence map, in order to avoid overstating the inferential scope of the survey design.

5.2 Reported preparedness gaps in global survey context

The Trust in AI student data reveal persistent gaps between rapid AI adoption and formal institutional scaffolding. Four in five students (83%) report using AI regularly in their studies, around half (53%) indicate that they have received some form of AI education or training, and approximately 72% report at least moderate AI knowledge and feel able to use AI tools effectively [39]. Students are more likely to use AI for study than employees are to use AI for work (83% versus 58%), indicating strong bottom-up adoption.

Provider-level support, however, appears comparatively limited. Only about half of students report that their institution offers appropriate safeguards, policies, training and resources for responsible AI use; fewer than one-third report the existence of an explicit GenAI policy; roughly one-fifth indicate that AI tools are banned; and about one-quarter do not know whether any policy exists [39]. Taken together, these survey results portray a global pattern in which students adopt general-purpose GenAI tools at scale while institutional policy, training and governance frameworks lag behind, a configuration that mirrors the policy lag and capacity deficits qualitatively documented in several of the reviewed higher-education studies. For example, Bittle and El-Gayar's systematic review of GenAI and academic integrity highlights the limited availability of clear institutional guidance around acceptable use in higher education, notwithstanding rapid uptake by students and staff [42], while Eğin et al. report low AI literacy and limited structured preparation among pre-service teachers [53].

5.3 Economic and infrastructure gradients as macro-context

The Trust in AI Report also documents pronounced capability gradients between countries, particularly when results are grouped by broad economic classification [39]. General populations in emerging economies report higher AI training exposure (approximately 50% versus less than 33%), higher self-reported AI knowledge (around 64% versus 46%), and higher AI efficacy (around 74% versus 51%) than respondents in advanced economies. These gradients do not permit conclusions about specific educational institutions or social groups, but they do suggest that AI capability is unevenly distributed across national and economic contexts, consistent with long-standing findings from multi-level digital divide research that emphasise both structural and usage-based inequalities [23–26].

Within the twelve-study corpus, infrastructure and economic barriers were identified as central constraints in Global South contexts. Boustani et al. detail how electricity shortages, limited fixed-line connectivity and high mobile data costs constrained

Lebanese students' ability to engage with GenAI and other digital tools [30]. Choi et al. show that in severely resource-constrained Sierra Leonean schools, questions of basic connectivity and device access must be resolved before large language models can be feasibly integrated into teaching practice [31]. In combination with such case-based evidence, the cross-national capability gradients reported in the Trust in AI study are best understood as macro-context: they indicate that structural resource and capability differences persist across economies, but do not by themselves license extrapolation about how GenAI is used within particular institutions or communities.

5.4 Digital literacy, capacity gaps and student-led adoption

The systematic review identified digital literacy and capacity gaps as significant barriers in five of the twelve studies. Eğin et al. found low AI literacy among both pre-service teachers and their educators, raising concerns that unequal preparation may widen existing divides if capacity-building is not prioritised [53]. Santiago-Ruiz, examining writing with ChatGPT in a context of educational inequality, similarly highlights how unequal access to digital skills and support can intersect with broader socio-economic disadvantage [28]. In mixed-methods work with engineering students in developing countries, Elkefi et al. report high levels of interest in and use of ChatGPT where access permits, but identify uneven skills and understanding of limitations, particularly among students with weaker prior digital experience [16].

The Trust in AI student module provides complementary evidence of such capacity gaps. Although a majority of students report regular AI use and moderate or higher self-assessed knowledge, only about half report having received structured AI training, and institutional support indicators lag behind usage [39]. Students overwhelmingly rely on free, general-purpose GenAI tools (approximately 89%), while only around one quarter use provider-supplied tools and a small minority use paid services [39]. This pattern structurally predisposes usage to outpace formal governance and training, because students' primary tools are not embedded in institutionally controlled platforms or curricula. The convergence between these survey patterns and the higher-education studies is therefore conceptual rather than statistical: both literatures point towards student-led adoption occurring under relatively thin institutional scaffolding, with uneven opportunities for systematic capacity development.

5.5 Integrative note on scope, strengths and limits

The Trust in AI Report analysis presented here should thus be read as an extension of the literature review, providing external descriptive context for the evidence map rather than an attempt to triangulate or generalise the twelve-study corpus to the population of all higher-education institutions. The principal strength of incorporating this contextual material lies in the convergence of themes across independent data sources and levels of analysis. Institutional preparedness deficits, policy lag and uneven capacity development appear both in education-sector studies, which report limited or fragmented guidance around GenAI use, and in cross-national survey indicators, which show that many students report substantial AI use alongside only partial institutional support [39, 42, 53]. Similarly, the persistence of capability gradients across economic classifications resonates with the structural and usage-based divides long documented in digital divide

scholarship [23–26] and with the infrastructural barriers observed in Lebanon and Sierra Leone [30, 31].

At the same time, the inferential limits are substantial. The Trust in AI student module does not provide country-resolved higher-education indicators that would permit rigorous alignment between national capability profiles and specific institutional contexts. Nor does it distinguish systematically between different types of higher-education provider or student subgroup. As a result, it is not possible to make claims about, for example, whether students in the particular Lebanese or Sierra Leonean institutions studied in the twelve-article corpus would fall above or below the global averages reported in the survey. The role of the global survey evidence in this article is therefore to illuminate the broader landscape in which GenAI–digital divide dynamics unfold, not to extend the empirical reach of the mapped studies.

The convergent patterns nonetheless have theoretical implications. They support digital divide frameworks that emphasise conversion factors, including institutional capacity, governance arrangements, infrastructure and pedagogical practices, as mediators between access, skills and meaningful outcomes. High general-population capability in several emerging economies coexisting with education-sector constraints, as seen in the Lebanese and Sierra Leonean cases, illustrates that access and basic literacy are necessary but not sufficient conditions for equitable GenAI integration. Equitable integration requires provider-side investments in staff time, assessment redesign, infrastructure reliability and inclusive policy processes. Recognising the Trust in AI Report as contextual rather than confirmatory evidence helps to avoid both techno-optimistic and techno-pessimistic readings, instead underscoring the need for sustained, empirically grounded capacity-building to ensure that GenAI serves as a bridge rather than a barrier in higher education.

6 Discussion

This systematised evidence map of twelve sources examining the intersection between GenAI adoption and digital divide phenomena in higher education reveals a complex and heterogeneous picture. Technological promise intersects with persistent structural inequalities, and the staged screening process that yielded only twelve relevant studies is itself indicative of the early and fragmented state of the evidence base. Given the small, non-random corpus and its uneven geographical coverage, the patterns identified here should be interpreted as indicative signals rather than representative estimates of sector-wide conditions.

Figure 4 provides a conceptual summary of how different categories of barriers recur across the included studies. It is based on qualitative coding of the twelve papers rather than on statistically representative sampling and is therefore intended as an analytical aid rather than a geographic or demographic map. Within this corpus, infrastructure barriers and digital literacy challenges are the most frequently discussed themes (appearing in seven and five studies respectively), while explicit analysis of policy interventions and their equity impacts is much less common (three studies). These frequencies describe the focus of published work rather than the prevalence of barriers in the wider higher-education sector.

Digital Divide Barriers in GenAI Adoption in Higher Education

Reporting frequency in mapped studies (n = 12): ■ High (≥7 studies) ■ Medium (4–6 studies) ■ Low (≤3 studies)

Barrier domain	Examples from resource-constrained contexts	Examples from better-resourced contexts	Reporting frequency	Illustrative studies
Infrastructure barriers (7 of 12 studies)	Electricity shortages Unreliable connectivity High mobile data costs Limited campus infrastructure	Platform and device constraints Integration challenges with LMS Local network or policy restrictions	High	Boustani et al. Choi et al. Elkefi et al. Hussein et al.
Digital literacy gaps (5 of 12 studies)	Low educator AI literacy Limited training access Basic technical skills only	Variable student proficiency Teacher readiness gaps Critical evaluation skills uneven	Medium	EğİN et al. Lin et al. Nguyen et al. Santiago-Ruiz
Economic barriers (5 of 12 studies)	High connectivity costs Device affordability issues Premium tool access limited	Premium feature costs Indirect access costs Computational resource needs	Medium	Boustani et al. Elkefi et al. Hussein et al.
Policy inadequacies (3 of 12 studies)	Limited guidance on GenAI Few equity-focused policies	Academic integrity focus Uneven institutional responses	Low	Bittle & El-Gayar Al Ghurair Foundation Hussein et al.

Note: Reporting frequency categories indicate how many of the 12 mapped studies discussed each barrier. They do not imply population-level prevalence, representativeness, or statistical effect size, and summarise patterns from case-based evidence only.

Fig. 4 Digital divide barriers matrix for GenAI adoption in higher education (conceptual summary based on the twelve included studies)

6.1 Theoretical framing and digital stratification mechanisms

The adoption patterns observed in the twelve studies can be interpreted through van Dijk's three-level digital divide framework, which distinguishes between access, skills and outcomes, and through subsequent refinements in educational contexts [24–26]. The first-level divide concerns material and infrastructural access. Evidence from Lebanon and Sierra Leone illustrates how unreliable electricity, limited bandwidth and unstable connectivity create barriers to meaningful engagement with GenAI tools [30, 31]. In the Lebanese case, affordability constraints, such as high mobile data costs and intermittent power, make GenAI tools functionally inaccessible for many students despite their nominal availability [30].

The second-level divide concerns digital and AI-related competencies. Several studies show that differences in technical proficiency and AI literacy shape whether students and educators are able to use GenAI productively [16, 53, 54]. EğİN et al. report low AI literacy among both pre-service teachers and their educators, noting that limited understanding of GenAI's capabilities and limitations risks inequitable integration [53]. Elkefi et al. identify uneven skills and awareness of risks among engineering students in developing countries [16], while Lin et al. observe substantial variation in students' ability to incorporate ChatGPT into online discussions in ways that support learning rather than substituting for it [54].

The third-level divide concerns differential outcomes from technology use. Here, the Trust in AI Report 2025 provides macro-level context rather than direct evidence on the specific institutions represented in the review. As discussed in Sect. 5, global student data from this report indicate high self-reported AI use and efficacy alongside only partial institutional support in terms of policies, training and safeguards [39]. These aggregate patterns suggest that, in many contexts, students' ability and willingness to use GenAI tools can outpace the institutional frameworks designed to support responsible and equitable use. Because the Trust in AI data are not resolved to the level of specific higher-education institutions or subgroups, they are used here solely as contextual background rather than as a basis for extrapolating from the twelve-study corpus.

Sen's capability approach further clarifies that access to resources alone is insufficient for achieving equitable outcomes. Students require conversion factors to transform GenAI access into valued educational functionings, including digital and AI literacy, institutional support, and appropriate pedagogical design [57]. Without these enabling conditions, the same tools can contribute to divergent outcomes, benefiting those who are already advantaged in terms of infrastructure, skills and institutional backing. Structural theories of cumulative advantage provide a useful heuristic here: when infrastructure, affordability and institutional support are uneven, groups who are already better positioned are more able to convert GenAI access into additional educational advantages, whereas others may be left behind.

Studies from Global South contexts in the corpus document grid instability, low-bandwidth networks and high connectivity costs that limit meaningful GenAI engagement for substantial segments of the student population [30, 31]. At the same time, work on bias in large language models shows that cultural and linguistic asymmetries in training data can impose additional cultural capital requirements for effective use, particularly for students from non-dominant linguistic and cultural backgrounds [32–34]. Taken together, these strands of evidence suggest that GenAI adoption is layered on top of existing digital and social stratifications, and that any equity-enhancing potential is contingent on how these underlying divides are addressed.

6.2 Adoption paradox and policy implications

A recurrent feature across several studies is an adoption paradox. Despite substantial infrastructural, economic and institutional constraints, students who do have access to GenAI often report high uptake and generally positive perceptions [6, 16, 54]. Elkefi et al. report very high levels of ChatGPT use among engineering students in developing countries where connectivity and device access are sufficient [16], while Nguyen et al. find that students in Southeast Asia frequently employ ChatGPT as part of their study routines [6]. These patterns suggest that perceived utility can be strong even in constrained environments.

Diffusion-of-innovation perspectives help to interpret this configuration: where students perceive clear relative advantages, compatibility with existing study practices and observable benefits (for example, faster drafting or access to explanations), adoption can proceed rapidly even when institutional scaffolding is limited. From this viewpoint, the global student data reported in the Trust in AI study can be read as contextual evidence of an efficacy–support asymmetry: students report high regular use and substantial self-assessed capability, yet only about half perceive provider policies, training, safeguards and resources, and many are unaware of whether a generative-AI policy exists [39]. These are contextual rather than sector-specific measures, but they are consistent with the pattern observed in the twelve-study corpus of student-led adoption under thin governance frameworks.

A digital-justice lens further clarifies the nature of the policy gaps. Justice-oriented accounts emphasise that redistribution (addressing material inequalities in connectivity, affordability and device access), recognition (addressing cultural and linguistic biases within GenAI systems and curricula) and representation (ensuring that diverse student and educator perspectives inform policy formation) are all needed if GenAI is to support rather than undermine educational equity. Within the small corpus reviewed

here, explicit evaluation of such interventions is largely absent. Many policy suggestions appear in discussion sections as normative recommendations (for example, calls for infrastructure investment or AI literacy programmes), but very few studies empirically examine the effectiveness of concrete interventions or their distributional consequences. This limits the strength of any claims that can be made about which policy approaches are most likely to work in different contexts.

6.3 Towards comprehensive frameworks for equitable implementation

The findings from this early evidence base challenge technologically deterministic narratives. GenAI does not inherently democratise or stratify education; rather, its effects are mediated by social, economic and institutional contexts. This observation aligns with socio-constructivist perspectives on technology, which emphasise that the meanings and consequences of tools are shaped by the practices, power relations and value frameworks into which they are introduced.

Critical and justice-oriented pedagogies offer a useful practical orientation. Several studies in the corpus point to the need for students to develop not only operational skills (for example, prompt formulation or summarisation) but also critical capacities to evaluate outputs, recognise biases and decide when not to use GenAI tools [16, 28, 53]. Effective integration therefore requires pedagogical approaches that value diverse ways of knowing, explicitly surface equity issues and support students to engage with GenAI as a contested socio-technical system rather than as a neutral assistant.

An equitable implementation framework, as suggested by the available evidence, would address material, cultural and pedagogical dimensions together. On the material side, infrastructure considerations must move beyond device counts to encompass reliability, affordability and sustainability of connectivity and power. On the capability side, AI literacy development should include critical evaluation, ethical reasoning and cultural awareness alongside technical competencies. On the institutional side, policy and governance processes would need to embed equity considerations into academic development, assessment practice and student support structures. Community and student engagement is also important, particularly in contexts where students are already using GenAI extensively without clear guidance.

At the same time, limits in the current evidence base remain significant. No longitudinal studies in the corpus track institutional preparedness and student outcomes over time, geographic coverage is uneven (with many regions unrepresented), and intervention evaluations are scarce. As a result, this review cannot provide strong claims about long-term effects or about the relative effectiveness of different implementation strategies. Rather, it highlights recurring problem configurations and points to areas where more robust empirical work is needed. To understand whether and under what conditions GenAI can contribute to narrowing rather than widening educational gaps, future research will need to move beyond documenting barriers towards systematically testing specific interventions and tracking their equity impacts over time.

7 Conclusion and future research

This systematised evidence map aimed to synthesise recent empirical work on the intersection of GenAI adoption and digital divide phenomena in higher education. Analysis of twelve studies published between 1 January 2023 and 30 January 2025 (including

online-first articles available by that date) suggests that whilst GenAI technologies such as ChatGPT possess substantial potential to support learning and productivity, their deployment is currently entangled with significant equity and access concerns. Within this limited corpus, the most frequently reported challenges include inadequate digital infrastructure, AI literacy deficits among students and educators, costs associated with access and usage, and the absence of comprehensive equity-focused institutional or national policies. These issues are especially visible in the small number of studies that examine Global South or otherwise marginalised contexts [16, 28, 30, 31], raising concerns that rapid GenAI diffusion could, under certain conditions, deepen existing educational disparities rather than ameliorate them.

7.1 Recommendations

Given the early stage of the evidence base, recommendations necessarily draw both on the findings of this review and on broader digital divide and digital equity scholarship [23, 24, 26]. National and international policymakers should consider sustained investment in accessible, secure and reliable digital infrastructure, particularly in underserved regions and communities, alongside the development of AI literacy frameworks that articulate minimum competencies for participation in GenAI-mediated learning. The reviewed studies and emerging bibliometric work on ChatGPT in education indicate that educational uses and challenges are evolving rapidly, underscoring the need for policy mechanisms that can adapt to new practices and risks [35]. Addressing bias in AI systems through systematic detection and mitigation strategies also appears critical for equitable implementation, especially where training data underrepresent particular languages, cultures or educational contexts [33, 36].

At the institutional level, universities and colleges are well placed to move from ad hoc responses towards more comprehensive institutional plans that explicitly integrate equity considerations into GenAI strategies. Potential elements include policies that promote equitable access to necessary hardware, software and connectivity; structured AI literacy and professional development programs for staff and students; and curriculum initiatives that embed critical engagement with GenAI's capabilities, limits and ethical implications. Local needs assessments can help institutions identify context-specific barriers (for example, campus-level connectivity constraints or particular student sub-groups at risk of exclusion) and target resources accordingly.

For educators, the reviewed studies highlight the importance of designing learning activities that acknowledge heterogeneous access and skills, and that support students to use GenAI reflectively rather than uncritically. This includes making explicit when and how GenAI tools can be appropriately used, discussing digital equity implications with students, and advocating within institutions for the resources required to support inclusive practice. Ongoing professional development in AI literacy and inclusive pedagogies is the most important enabling condition, although empirical evidence on which models are most effective is not yet available.

7.2 Limitations

Several limitations of this review warrant reiteration. First, the temporal restriction to 1 January 2023 through 30 January 2025 meant that only studies published in the immediate post-ChatGPT period were included. This captured the most recent wave of

GenAI-related work but may have excluded earlier research on digital equity in educational technology that could offer relevant insights. Second, the search strategy, although multi-stage and conducted across several databases plus targeted grey-literature sources, was limited to English-language publications and thus may have missed studies in other languages, particularly from regions where digital divide issues are most acute.

Third, the small corpus of twelve studies, with uneven geographical distribution and a predominance of cross-sectional designs, constrains generalisability. The review does not provide a representative picture of global higher education; rather, it collates evidence from a set of early empirical contributions that explicitly engage with both GenAI and digital divide themes. Fourth, grey literature inclusion was limited to a small number of foundation and institutional reports with identifiable empirical content, and it is possible that additional relevant reports were not captured. Finally, while the thematic analysis was conducted systematically and discussed within the research team, it remains interpretive and is conditioned by the available primary studies.

7.3 Directions for future research

The nascent and fragmented nature of the empirical literature points to a clear shift in research priorities. The barriers identified in this review, relating to infrastructure, literacy, policy and affordability, are relatively consistent across the small number of contexts studied, but there is very limited evidence on how these barriers can be effectively mitigated in practice. Future work therefore needs to move beyond problem description towards the design and evaluation of targeted interventions.

7.3.1 Intervention design and evaluation

A primary priority is rigorous evaluation of specific pedagogical, infrastructural and policy interventions aimed at supporting equitable GenAI integration. Possible approaches include controlled or quasi-experimental studies of AI literacy curricula, evaluations of institutional policy packages that combine access support with clear usage guidelines, and trials of targeted infrastructure support (for example, subsidised connectivity or loan-device schemes) in combination with GenAI-enabled learning designs. Study designs such as cluster randomised trials, stepped-wedge designs or quasi-experiments with matched comparison groups could be employed where feasible, with pre-registered outcomes and explicit attention to differential effects across student subgroups. Unlike earlier educational technologies where access barriers were primarily device-based, GenAI interventions must address distinctive equity dimensions. These include: (a) subscription-tier economics, requiring evaluation of differential learning outcomes between users of free versus premium services; (b) linguistic and cultural bias in model outputs, necessitating culturally-responsive literacy frameworks that enable critical evaluation of algorithmically generated content; and (c) compute-intensity requirements for advanced features (image generation, extended context, real-time web access), which compound infrastructure barriers in low-bandwidth contexts. Future intervention studies should explicitly operationalise these GenAI-specific dimensions rather than treating GenAI as generic educational technology.

7.3.2 Secondary research priorities

Several complementary priorities emerge from the gaps identified in the evidence map. Geographic diversification is essential: empirical investigations from underrepresented regions in Africa, Latin America, the Middle East and large parts of Asia are needed to capture the diversity of infrastructural and institutional contexts in which GenAI is being deployed. Longitudinal research is also crucial, as existing studies largely capture early adoption stages rather than long-term dynamics. Tracking how GenAI use, institutional preparedness and equity gaps evolve over time would help to distinguish transient adjustment effects from more durable patterns.

Further, there is a need for systematic comparative work on pedagogical integration strategies. Different disciplines and programme types may require different forms of GenAI use (for example, coding assistance versus writing support), and equity impacts may vary accordingly. Policy analysis represents another important area, including comparative studies of national and institutional regulatory frameworks and their effects on access, usage patterns and perceived fairness.

Finally, the development and validation of higher-education-specific AI literacy frameworks that explicitly integrate equity considerations would provide a more coherent foundation for curriculum and policy design. This includes not only technical competencies but also critical evaluation skills, understanding of bias and transparency issues, and awareness of how GenAI systems are embedded in wider socio-economic inequalities. Research focused on marginalised student populations, for example, by socioeconomic status, disability, regional location or migrant background, also requires expansion, as existing studies provide only limited insight into how GenAI-mediated learning is experienced by those most at risk of exclusion.

Overall, the present review highlights that questions of digital justice and equity are central to understanding the educational implications of GenAI. While the current evidence base is too limited to support broad generalisations, it already indicates that without deliberate, empirically informed interventions, GenAI is unlikely on its own to reduce existing digital divides in higher education and may, in some contexts, contribute to new forms of stratification.

7.4 Use of AI tools

In accordance with Springer's guidelines, the authors declare that generative artificial intelligence tools (Claude and ChatGPT) were employed to assist with manuscript preparation and revision for editorial and organisational improvements. These tools were used solely to enhance readability, consolidate subsections, and improve manuscript structure and flow. All research design, data collection, analysis, interpretation of results, and substantive content creation were conducted by the authors. The authors take full responsibility for the accuracy, integrity, and originality of the research findings and conclusions presented in this article.

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Author contributions

R.J.E. conceived the study, designed the methodology, conducted the systematic literature search and screening, performed the formal analysis and thematic synthesis, and drafted the original manuscript. E.G. contributed to study conceptualisation, provided supervision throughout the research process, and participated in manuscript review and editing. A.A.-S. contributed to study conceptualisation and methodology design, provided supervision, and participated

in manuscript review and editing. R.J.E. managed project administration and coordinated the research activities. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

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Consent for publication

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Competing of interests

The authors declare no competing interests.

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References

1. Sandu R, Gide E, Elkhodr M. The role and impact of chatgpt in educational practices: insights from an Australian higher education case study. *Discover Educ.* 2024;3(1):71. <https://doi.org/10.1007/s44217-024-00126-6>.
2. Yu H. The application and challenges of chatgpt in educational transformation: New demands for teachers' roles. *Heliyon.* 2024;10(2).
3. Chiu TKF. The impact of generative ai (genai) on practices, policies and research direction in education: A case of chatgpt and midjourney. *Interact Learn Environ.* 2024;32(10):6187–203.
4. Bernabei M, Colabianchi S, Falegnami A, Costantino F. Students' use of large language models in engineering education: a case study on technology acceptance, perceptions, efficacy, and detection chances. *Comput Educ Artif Intell.* 2023;5:100172.
5. Mohamed FN, Azhar J, Yasmien S, Hussain I, Khawar M. Generative artificial intelligence and personalized learning environment: challenges and opportunities. *Southern J Comput Sci.* 2025;1(01):1–36.
6. Nguyen TNT, Van Lai N, Nguyen QT. Artificial intelligence (ai) in education: a case study on chatgpt's influence on student learning behaviors. *Educ Process Int J.* 2024;13(2):105–21.
7. Jayaweera I. Revolutionizing education: generative ai as a catalyst for personalized learning and innovative teaching practices. In: 2025 5th international conference on advanced research in computing (ICARC), IEEE 2025;1–6.
8. Sharma S, Mittal P, Kumar M, Bhardwaj V. The role of large language models in personalized learning: a systematic review of educational impact. *Discover Sustain.* 2025;6(1):1–24.
9. Wang S, Xu T, Li H, Zhang C, Liang J, Tang J, Yu PS, Wen Q. Large language models for education: a survey and outlook;2024 arXiv preprint [arXiv:2403.18105](https://arxiv.org/abs/2403.18105)
10. Abdaljeel M, Barakat M, Alsanafi M, Salim NA, Abazid H, Malaeb D, et al. A multinational study on the factors influencing university students' attitudes and usage of chatgpt. *Sci Rep.* 2024;14(1):1983.
11. Bello SB, Suleiman ZA, Abdurrahman MS, Bashar A, Ibrahim JA. Perceptions of ease of use and usefulness of artificial intelligence chatgpt among undergraduate law students of Ahmadu bello university, Zaria;2024.
12. Shoufan A. Exploring students' perceptions of chatgpt: thematic analysis and follow-up survey. *IEEE Access.* 2023;11:38805–18. <https://doi.org/10.1109/ACCESS.2023.3268224>.
13. Tiwari CK, Bhat MA, Khan ST, Subramaniam R, Khan MAI. What drives students toward chatgpt? An investigation of the factors influencing adoption and usage of chatgpt. *Interact Technol Smart Educ.* 2024;21(3):333–55.
14. Youssef E, Medhat M, Abdellatif S, Al Malek M. Examining the effect of chatgpt usage on students' academic learning and achievement: a survey-based study in ajman, uae. *Comput Educ Artif Intell.* 2024;7:100316.
15. Kamoun F, El Ayeb W, Jabri I, Sifi S, Iqbal F. Exploring students' and faculty's knowledge, attitudes, and perceptions towards chatgpt: a cross-sectional empirical study. *J Inform Technol Educ Res.* 2024;23.
16. Elkefi S, Tounsi A, Kefi MA. Use of chatgpt for education by engineering students in developing countries: a mixed-methods study. *Behav Inform Technol.* 2024;1–17.
17. Bera S, Vilchez A, Muenster RM. The persistent global disparities in environmental and climate communication scholarship. *Frontiers Commun.* 2025;9:1474619.
18. Egamberdiyeva N. Gender equality in education: global perspectives and best practices. *Mental Enlight Sci Methodol J.* 2025;6(01):48–56.
19. Hennekam S, Kulkarni M, Beatty JE. Neurodivergence and the persistence of neurotypical norms and inequalities in educational and occupational settings. *Work Employ Soc.* 2025;39(2):449–69.
20. Muyambi GC, Ahiaku PKA. Inequalities and education in South Africa: a scoping review. *Int J Educ Res Open.* 2025;8:100408.
21. Assefa Y, Gebremeskel MM, Moges BT, Tilwani SA, Azmera YA. Rethinking the digital divide and associated educational in (equity) in higher education in the context of developing countries: the social justice perspective. *Int J Inform Learn Technol.* 2025;42(1):15–32.
22. Stepnov IM, Kasatkin PI, Kolesnikova TV. Digital divide: Understanding the disparities between global north and global south. In: *Polycrisis and economic development in the global South*, Routledge, London, 2024:171–183.
23. Hargittai E, Hinnant A. Digital inequality: differences in young adults' use of the internet. *Commun Res.* 2008;35(5):602–21.

24. Van Deursen AJ, Van Dijk JA. The digital divide shifts to differences in usage. *New Med Soc*. 2014;16(3):507–26.
25. Van Deursen AJ, Helsper EJ. The third-level digital divide: Who benefits most from being online? In: *Communication and Information Technologies Annual*. Emerald Group Publishing Limited Bingley. 2015;10:29–52. <https://doi.org/10.1108/S2050-206020150000010002>.
26. Martin F, Ceviker E, Gezer T. From digital divide to digital equity: systematic review of two decades of research on educational digital divide factors, dimensions, and interventions. *J Res Technol Educ*. 2024. <https://doi.org/10.1080/15391523.2024.2425442>.
27. Vitalis PO, Aondover EM, Ogunbola O, Onyejelem TE, Ridwan M. Accessing digital divide and implications in Nigeria: the media dimension. *Budapest Int Res Critics Inst J (BIRCI-Journal)* 2025;8(1):1–12.
28. Santiago-Ruiz E. Writing with chatgpt in a context of educational inequality and digital divide. *Int J Educ Dev Using Inf Commun Technol*. 2023;19(3):28–38.
29. Zipf S, Wu C, Petricini T. Using the information inequity framework to study genai equity: analysis of educational perspectives. *Inform Res Int Electron J*. 2025;30:533–47.
30. Boustani NM, Sidani D, Boustany Z. Leveraging ict and generative ai in higher education for sustainable development: the case of a lebanese private university. *Adm Sci*. 2024;14(10):251.
31. Choi JH, Garrod O, Atherton P, Joyce-Gibbons A, Mason-Sesay M, Björkegren D. Are llms useful in the poorest schools? Theteacher.ai in sierra leone;2023 arXiv preprint [arXiv:2310.02982](https://arxiv.org/abs/2310.02982)
32. Dempere J, Modugu K, Hesham A, Ramasamy LK. The impact of chatgpt on higher education. *Frontiers Educ*. 2023;8:1206936. <https://doi.org/10.3389/educ.2023.1206936>.
33. Lin L, Wang L, Guo J, Wong K-F. Investigating bias in llm-based bias detection: disparities between llms and human perception;2024. arXiv preprint [arXiv:2403.14896](https://arxiv.org/abs/2403.14896)
34. Sallam M. Chatgpt utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. *Healthcare*. 2023;11(6):887. <https://doi.org/10.3390/healthcare11060887>.
35. Samala AD, Sokolova EV, Grassini S, Rawas S. Chatgpt: a bibliometric analysis and visualization of emerging educational trends, challenges, and applications. *Int J Eval Res Educ (IJERE)*. 2024;13(4):2374–87. <https://doi.org/10.11591/ijere.v13i4.28119>.
36. Samala AD, Rawas S. Bias in artificial intelligence: smart solutions for detection, mitigation, and ethical strategies in real-world applications. *Int J Artif Intell*. 2025;2252(8938):8938.
37. Hine E, Floridi L. Artificial intelligence with American values and Chinese characteristics: a comparative analysis of American and Chinese governmental ai policies. *AI Soc*. 2024;39(1):257–78.
38. Sallam M, Al-Mahzoum K, Sallam M, Mijwil MM. Deepseek: Is it the end of generative ai monopoly or the mark of the impending doomsday? *Mesop J Big Data*. 2025;2025:26–34.
39. Gillespie N, Lockey S, Ward T, Macdade A, Hassed G. Trust, attitudes and use of artificial intelligence: a global study 2025. Technical report, The University of Melbourne and KPMG;2025. <https://doi.org/10.26188/28822919>. https://mbs.edu/-/media/PDF/Research/Trust_in_AI_Report.pdf
40. McDonald N, Johri A, Ali A, Collier AH. Generative artificial intelligence in higher education: evidence from an analysis of institutional policies and guidelines. *Comput Human Behav Artif Humans*. 2025;3:100121.
41. Wang H, Dang A, Wu Z, Mac S. Generative ai in higher education: seeing chatgpt through universities' policies, resources, and guidelines. *Comput Educ Artif Intell*. 2024;7:100326.
42. Bittle K, El-Gayar O. Generative ai and academic integrity in higher education: a systematic review and research agenda. *Information*. 2025;16(4):296.
43. Hussein H, Gordon M, Hodgkinson C, Foreman R, Wagad S. Chatgpt's impact across sectors: a systematic review of key themes and challenges. *Big Data Cognitive Comput*. 2025;9(3):56.
44. Bond M, Khosravi H, De Laat M, Bergdahl N, Negrea V, Oxley E, et al. A meta systematic review of artificial intelligence in higher education: a call for increased ethics, collaboration, and rigour. *Int J Educ Technol High Educ*. 2024;21(1):4.
45. Zawacki-Richter O, Marín VI, Bond M, Gouverneur F. Systematic review of research on artificial intelligence applications in higher education-where are the educators? *Int J Educ Technol High Educ*. 2019;16(1):1–27.
46. Batista J, Mesquita A, Carnaz G. Generative ai and higher education: trends, challenges, and future directions from a systematic literature review. *Information*. 2024;15(11):676.
47. Wang X, Zainuddin Z, Chin HL. Generative artificial intelligence in pedagogical practices: a systematic review of empirical studies (2022–2024). *Cogent Educ*. 2025;12(1):2485499.
48. George B, Mattathil AP. Empowering African American tourism entrepreneurs with generative ai: bridging innovation and cultural heritage. *Societies*. 2025;15(2):34.
49. Foundation AAG. Ignite techtalk lebanon: Empowering youth in Lebanon through digital upskilling and jobs. Technical report;2023.
50. Munn Z, Peters MDJ, Stern C, Tufanaru C, McArthur A, Aromataris E. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Med Res Methodol*. 2018;18:1–7.
51. ...Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. Prisma extension for scoping reviews (prisma-scr): checklist and explanation. *Ann Intern Med*. 2018;169(7):467–73. <https://doi.org/10.7326/M18-0850>.
52. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372.
53. Eğin F, Onan A, Yildiz Durak H. Let's talk about edtech! a topic modelling analysis of ai tools and pre-service teachers' perspectives. *Eur J Educ*. 2025;60(1):12913. <https://doi.org/10.1111/ejed.12913>.
54. Lin X, Luterbach K, Gregory KH, Sconyers SE. A case study investigating the utilization of chatgpt in online discussions. *Online Learn*. 2024;28(2):2.
55. Rogers EM. *Diffusion of innovations*. 5th ed. New York City, NY: Simon and Schuster; 2003.
56. Beisly A, Abeyrathna Herath Mudiyanseleg D. Teachers can use it with their lillies: using case studies to explore preservice teachers' perceptions of technology. *Educ Sci*. 2025;15(3):366.

57. Sen A. Development as freedom oxford University Press shaw tm & heard. The Politics of Africa: Dependence and Development; 1999.

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